# Acute myeloid leukemia

**≥20% myeloid blasts — or any blast % with a defining genetic lesion** · classification and treatment are now genetically driven · every new diagnosis is triaged for the three emergencies before a regimen is chosen

*Compiled from the Encyclopedia AML review note and Fellowship Anki deck · ELN 2022 · WHO 2022 / ICC · NCCN.*

## DIAGNOSIS

- **≥20% myeloid blasts — or any blast % with a defining fusion**: t(8;21), inv(16)/t(16;16), PML::RARA. WHO 2022 also drops it for NPM1 but **keeps ≥20% for CEBPA-bZIP**; **ICC sets ≥10% for both**
- Also AML at <20%: **myeloid sarcoma** / extramedullary disease
- "Blasts" on smear is morphologic — **confirm lineage by flow**
- Workup: marrow aspirate + biopsy with **flow, karyotype + FISH, NGS**; CBC/smear (Auer rods); **coags + fibrinogen + D-dimer**; TLS labs; **baseline echo**; HLA typing early; LP only for CNS symptoms (consider if monocytic), after coags corrected
- **Stat FISH/PCR for PML::RARA** on any suspicion of APL
- Median age ~68. The three triage questions on every new AML: **Is it APL? Is there leukostasis? Is there tumor lysis?**

## ELN 2022 GENETIC RISK

| Risk | Representative genetics |
|---|---|
| **Favorable** | t(8;21) RUNX1::RUNX1T1 · inv(16)/t(16;16) CBFB::MYH11 (**core-binding factor**) · **mutated NPM1 without FLT3-ITD** · in-frame bZIP CEBPA |
| **Intermediate** | NPM1 + FLT3-ITD · wild-type NPM1 + FLT3-ITD · t(9;11) MLLT3::KMT2A · anything not otherwise classified |
| **Adverse** | Complex or monosomal karyotype · –5/del(5q), –7, abn(17p) · **TP53** · MDS-related genes (ASXL1, BCOR, EZH2, RUNX1, SF3B1, SRSF2, STAG2, U2AF1, ZRSR2) · KMT2A rearrangements **except t(9;11) and KMT2A-PTD** · **t(6;9), inv(3)** |

**ELN 2022 changes worth knowing:** FLT3-ITD is intermediate **regardless of allelic ratio** (ratio dropped) · only **in-frame bZIP** CEBPA is favorable · **NPM1 with adverse cytogenetics is reclassified adverse** · **KIT co-mutation worsens CBF-AML prognostically, but ELN 2022 kept it favorable**.

## GENOTYPE → PHENOTYPE

| | |
|---|---|
| **FLT3-ITD** | Hyperleukocytosis; normal karyotype. ↑ **CNS risk**, most when **NPM1 co-mutated** |
| **KMT2A / 11q23** | Monocytic — gum hypertrophy, leukemia cutis. **Highest multivariable odds for CNS disease** (OR ~7.6) |
| **t(8;21)** | **Myeloid sarcoma**; aberrant **CD19** / **CD56** by flow |
| **inv(16)** | **Abnormal marrow eosinophils** with coarse basophilic granules (FAB M4Eo) |
| **TP53** | Complex karyotype, t-AML; **venetoclax-resistant**, poor outcome even after allo |
| **DTA** | **DNMT3A, TET2, ASXL1 persist in CR** as clonal hematopoiesis — **not MRD markers** |

CNS disease is uncommon — **~0.6% at diagnosis, ~3% at relapse**. Risk factors differ by series: SAL (n=3,261) found **FLT3-ITD ± NPM1**; another found **11q23 (OR 7.6)** and **WBC ≥100k** on multivariable analysis.

## TREATMENT PATHWAY

**First: exclude APL, then assess fitness**

### Before committing to a regimen

**APL must be excluded** — ATRA cannot wait
7+3 may start while NGS pends, but **FLT3** status changes induction — hold a day or two if stable; instability forces faster action

**Fit for intensive therapy**

### Induction — 7+3

cytarabine ×7 days + anthracycline ×3
**FLT3-mutated: add midostaurin** (RATIFY)
**CD33 / CBF: add gemtuzumab** (ALFA-0701)
secondary or t-AML: consider CPX-351

### PARADIGM (2026)

**adverse risk and not FLT3-mutated**: aza+ven **EFS 14.5 vs 6.2 mo**, HR 0.57. Less infection and bleeding, more reached allo · **phase 2; OS NS**

### Assess response

**day-14 marrow**; residual disease → re-induce
goal is CR: blasts <5% with count recovery

### Consolidation by ELN risk

**favorable** → HiDAC ×3–4 cycles, no allo
**intermediate / adverse → allo-HSCT in CR1**
MRD-positive CR pushes toward transplant

### If not going to transplant

**oral azacitidine (CC-486) maintenance**
QUAZAR: OS 24.7 vs 14.8 months

**Unfit for intensive therapy**

### Venetoclax + azacitidine

**VIALE-A**: OS **14.7 vs 9.6 months** vs aza alone
**watch TLS on ramp-up**; reduce dose with azoles
IDH1-mutant alternative: ivosidenib + aza (AGILE)

**Relapsed / refractory**

### Target first, then salvage

**FLT3** → gilteritinib (ADMIRAL)
**IDH1/2** → ivosidenib / enasidenib
**KMT2A-r or NPM1** → menin inhibitor
salvage chemo: FLAG-IDA, MEC, CLAG-M

## REGIMENS & AGENTS

| Regimen | Notes |
|---|---|
| **7+3** | Cytarabine infusion ×7 days + an anthracycline ×3 — the backbone for fit patients. **Daunorubicin 60 mg/m² is the modern standard:** E1900 showed **90 > 45**, but **AML17 found 90 vs 60 gave no CR or OS benefit and doubled 60-day mortality** |
| **+ midostaurin** | FLT3-mutated (RATIFY). Quizartinib for FLT3-ITD (QuANTUM-First) |
| **+ gemtuzumab** | Fractionated dosing (ALFA-0701 — no CD33 cutoff). **CBF benefit comes from the Hills meta-analysis**, not ALFA-0701 |
| **CPX-351** | Liposomal cytarabine + daunorubicin for t-AML / AML-MRC. **Caveat:** comparator was 7+3, not ven/aza — niche today is the fitter older adult bridging to allo |
| **HiDAC** | 3 g/m² q12h **on days 1, 3, 5** ×3–4 cycles; best in CBF. **Age >60 → IDAC 1.5 g/m²** — 3 g/m² is inappropriate (ocular, cerebellar, infectious toxicity). Renal dose-reduce |
| **Venetoclax + azacitidine** | Unfit patients (VIALE-A). Ramp-up with TLS prophylaxis; **reduce venetoclax with CYP3A inhibitors** |

This table is **when and why**. **Every regimen's dosing lives on page 3** — the two HiDAC numbers stay here because the dose *is* the teaching point.

## MRD AND THE TRANSPLANT DECISION

- **Measure MRD at the end of every cycle** and again before transplant — order **HemeSTAMP MRD** (NGS) **+ multiparameter flow cytometry**; add **NPM1 PCR** where applicable
- **MRD-positive CR is not CR for planning purposes** — it predicts relapse and moves an otherwise favorable-risk patient toward transplant
- **Allo-HSCT in CR1:** generally **not** for favorable risk · **yes** for intermediate and adverse risk · modified by MRD, age, comorbidity, and donor availability
- Allo cures through **graft-versus-leukemia**. **Auto-HSCT lacks GVL** and relapses more — niche for MRD-negative CR1 without a donor
- Fitness for intensive therapy is a composite of age, performance status and comorbidity (HCT-CI) — not age alone

# Acute myeloid leukemia



## EMERGENCIES

### APL

**Recognize**
DIC + bleeding + hypergranular promyelocytes
**bundled Auer rods**
microgranular variant can look monocytic

**Start ATRA immediately**
do **not** wait for PML::RARA — hemorrhage is the leading cause of APL death, and it is preventable
**45 mg/m²/day PO divided q12h** (22.5 mg/m² BID); **a one-time dose overnight is fine**

**Support aggressively**
**fibrinogen >100** (cryo), **platelets >30k**, **INR <2**
coags q6–8 h; avoid LP and central lines

**Risk-stratify, then treat**
**Sanz:** low = WBC ≤10k *and* plt >40k · intermediate = WBC ≤10k *and* plt ≤40k · **high = WBC >10k**
**ATRA + arsenic**, chemo-free, for low/intermediate
**high risk: add anthracycline** (or GO)
cure >90%; **do NOT leukapherese — worsens DIC**

### Leukostasis

**Hyperleukocytosis + symptoms**
myeloid blasts are large and sticky — symptoms at lower counts than ALL/CLL, often >50–100k
**CNS** (headache, confusion, vision) + **pulmonary** (hypoxia)

**Management — with doses**
**hydroxyurea up to 2 g PO q6h**, goal **WBC <20k**
literature 50–100 mg/kg/day; **mucositis ≥75 mg/kg/day**
HU takes days — if too slow, **cytarabine 1 g ×1**
**urgent induction is definitive**; **leukapheresis if severe** — **never in APL**, no mortality benefit. **No RBC transfusion**
hydrate + TLS ppx. **dex 10 mg IV q6h** if monocytic + ARF

### Tumor lysis

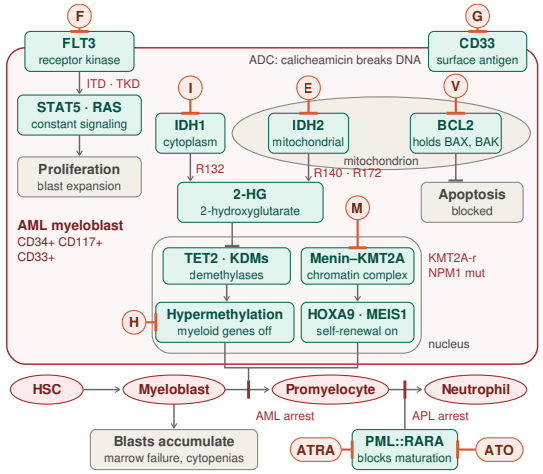
**Cairo-Bishop**
lab TLS = ≥2 of ↑K, ↑phosphate, ↑uric acid, ↓Ca
clinical TLS = lab TLS + AKI, arrhythmia or seizure

**Prevent and treat**
**aggressive IV hydration** is the cornerstone
allopurinol 300 mg BID standard risk
**rasburicase** if high risk or uric acid >8 — **not in G6PD**
**no urinary alkalinization**

## DRUG TOXICITY & MONITORING

| Agent | Watch for |
|---|---|
| **Cytarabine** | Pyrimidine analog. High-dose: **cerebellar toxicity — cerebellar exam before every HiDAC dose, hold if abnormal**; **conjunctivitis → prophylactic steroid eye drops**; ara-C syndrome (fever, myalgia, rash) |
| **Anthracyclines** | Cumulative, dose-dependent **cardiotoxicity** — track total dose, baseline ± serial LVEF. **Vesicant** — central access |
| **Venetoclax** | **Tumor lysis on ramp-up**; neutropenia; **dose-reduce with azole antifungals** (CYP3A) |
| **Azacitidine vs decitabine** | Both hypomethylating; **azacitidine incorporates into RNA and DNA**, decitabine DNA-only. Azacitidine: injection-site reactions |
| **Gemtuzumab** | Anti-CD33 calicheamicin ADC — **hepatotoxicity and VOD/SOS**, especially peri-transplant; fractionated dosing lowers risk |
| **Arsenic trioxide** | **QT prolongation** — monitor ECG, replete K and Mg |
| **Differentiation syndrome** | ATRA/ATO **and also IDH and menin inhibitors**. Fever, dyspnea, hypoxia, infiltrates, effusions, weight gain, AKI → **dexamethasone 10 mg IV BID at first suspicion** |
| **Menin inhibitors** | Revumenib, ziftomenib — differentiation syndrome + **QT prolongation** |

## PATHOPHYSIOLOGY




(F) midostaurin · gilteritinib (type I: ITD + TKD) · quizartinib (type II: ITD only)
(I) ivosidenib: IDH1 inhibitor
(E) enasidenib: IDH2 inhibitor
(M) revumenib · ziftomenib: menin inhibitors
(V) venetoclax: BCL2 inhibitor (+ HMA)
(H) azacitidine · decitabine: DNMT inhibitors
(G) gemtuzumab ozogamicin: CD33 ADC
(ATRA) binds RARA (ATO) arsenic trioxide: binds PML; both degrade PML::RARA
Differentiation syndrome risk: ATRA/ATO, IDH inhibitors, menin inhibitors, gilteritinib.
Quizartinib binds only inactive FLT3, so TKD mutations escape it; gilteritinib hits both.

## TRIAL EVIDENCE

| | |
|---|---|
| **RATIFY** | Midostaurin + 7+3 in FLT3-mutated — survival benefit |
| **QuANTUM-First** | Quizartinib + 7+3 in FLT3-ITD |
| **ALFA-0701** | Fractionated gemtuzumab + 7+3 — EFS/RFS benefit (OS NS at final analysis). **Hills meta-analysis** (n=3,325): **favorable-cytogenetics 5-yr OS +20.7%** |
| **E1900** | Daunorubicin **90 vs 45 mg/m²** — better CR and OS in age <60 |
| **AML17** | Daunorubicin **90 vs 60 mg/m²** — **no CR or OS difference**, and 60-day mortality **10% vs 5%** (HR 1.98). **Read with E1900: 60 is standard, not 90** (Burnett, Blood 2015) |
| **VIALE-A** | Azacitidine + venetoclax vs azacitidine — OS **14.7 vs 9.6 months** (DiNardo, NEJM 2020) |
| **PARADIGM** | **Aza+ven vs intensive induction in *fit* patients** (CBF/FLT3/NPM1<60 excluded, 72% ELN adverse): EFS **14.5 vs 6.2 mo**, HR 0.57; less infection and bleeding. Phase 2; OS NS (Fathi, NEJM 2026) |
| **AGILE** | Ivosidenib + azacitidine in IDH1-mutant unfit patients |
| **ADMIRAL** | Gilteritinib in relapsed FLT3-mutated AML — survival benefit |
| **SORMAIN** | **Sorafenib maintenance after allo** in FLT3-ITD AML — 24-mo RFS **85% vs 53%**, HR 0.39. Phase 2, n=83; **off-label** (Burchert, JCO 2020) |
| **MORPHO** | **Gilteritinib maintenance after allo** in FLT3-ITD — RFS NS overall; **benefit only if MRD-positive** pre- or post-HCT (HR 0.52). **An MRD-guided decision** (Levis, JCO 2024) |
| **APL0406** | ATRA + arsenic beat ATRA + chemo in low/intermediate-risk APL (Lo-Coco, NEJM 2013) |
| **QUAZAR AML-001** | Oral azacitidine (CC-486) maintenance in CR1 — OS **24.7 vs 14.8 months** |

## GENETICS — ADDITIONAL ASSOCIATIONS

| | |
|---|---|
| **inv(3) / t(3;3)** | MECOM (EVI1, 3q26) activation; classically **normal or high platelets**; adverse and **venetoclax-resistant** |
| **t(6;9)** | DEK::NUP214 — associates with **FLT3-ITD and basophilia**; adverse |
| **NPM1 flow mimic** | NPM1-mutated AML can be **HLA-DR–** / **CD34–** and mimic APL — confirm PML::RARA |
| **Therapy-related** | After **alkylators/RT**: del(5q)/del(7q) ± TP53, latency ~5–7 yr. After **topoisomerase-II**: **11q23 KMT2A**, latency ~2–3 yr |
| **Germline** | **DDX41, RUNX1, CEBPA, GATA2**, telomere biology — send germline testing if young, family history, or multiple cytopenias; **it changes related-donor selection** |

# Acute myeloid leukemia



## FOUR WAYS AN ACUTE LEUKEMIC DIES IN THE ED

- **1. Leukostasis** — CVA, MI, respiratory failure
- **2. Bleeding** / **DIC** / **thrombosis**
- **3. Infection** — **very low threshold for broad-spectrum antibiotics**
- **4. Tumor lysis** — renal failure, electrolyte derangement

## OVERNIGHT DECISIONS

- **1. Triage floor vs ICU** — **leukapheresis and CRRT both require ICU**
- **2. Rule out APL** — **start ATRA on any suspicion**, do not wait for PML::RARA
- **3. Cytoreduction** — see doses at right
- **4. TLS** — start fluids + allopurinol, set lab frequency (**q4h to daily**). **Rasburicase if uric acid >8: 0.15–0.2 mg/kg IV**, in adults usually a **single dose capped at 6 mg** (label allows 0.2 mg/kg daily ×5 days). **Check G6PD — contraindicated if deficient**

## SUPPORTIVE CARE

- **Transfusion:** platelets >10k (>20–50k if bleeding, procedure, or APL); **irradiated and leukoreduced** products
- **Prophylaxis during induction: levofloxacin 500 mg PO daily** · **posaconazole DR 300 mg BID ×1 day, then 300 mg daily** · **acyclovir 400 mg PO BID** (HSV-seropositive)
- **Posaconazole is a strong CYP3A4 inhibitor** — that is what forces the venetoclax reduction: **400 mg → 70 mg daily** with posaconazole (100 mg with other strong inhibitors such as voriconazole), during ramp-up and beyond
- **Hepatitis B:** screen HBsAg + anti-HBc before therapy. **HBsAg-positive → entecavir or tenofovir** (preferred over lamivudine) for **any** systemic anticancer therapy, continued **≥12 months** past the last cycle. Isolated anti-HBc-positive → prophylax or monitor HBV DNA closely; AML chemotherapy carries reactivation risk similar to lymphoma
- **Febrile neutropenia:** fever ≥38.3°C once (or ≥38.0°C for 1 h) with ANC <500 → **cefepime 2 g IV q8h within 1 hour**. **Add vancomycin** for hemodynamic instability, line infection, skin/soft tissue infection, pneumonia or known MRSA. **Severe mucositis is a softer trigger than it looks** — IDSA lists it only when fluoroquinolone prophylaxis was used *and* ceftazidime is the empiric agent (viridans strep); **cefepime already covers viridans strep**. Antifungal if fever persists >4–7 days
- Fertility counseling before chemotherapy; central venous access; early palliative care in older or adverse-risk disease

## WHAT TO SEND

| Test | When |
|---|---|
| **Peripheral studies** | **Flow cytometry**, **AML prognosis assay** (NPM1, FLT3-ITD, FLT3-TKD), **IDH mutation analysis**. Send if **WBC >10k** and marrow will be delayed, or the patient is old enough that you would rather not subject them to a marrow |
| **Cytogenetics** | **FISH for PML::RARA and BCR::ABL1**. Send **from peripheral blood** if there is any concern for APL or a delay in getting marrow |
| **HLA typing** | Send if the patient is a possible allogeneic transplant candidate |

## CYTOREDUCTION OVERNIGHT

| | Dose and trigger |
|---|---|
| **Hydroxyurea** | Consider if **WBC >20k**. Overnight you can order a **one-time dose**. Typical range **500 mg–2 g**, **max 2 g q6h**. Literature range 50–100 mg/kg/day; **grade 3–4 mucositis rises at ≥75 mg/kg/day**, so 2 g q6h is the ceiling, not the starting dose |
| **Leukapheresis** | Consider for **evidence of leukostasis** — headache, blurry vision, dyspnea/hypoxia, chest pain — typically with **blasts >50k in AML** (rare in ALL). **Page apheresis 12027.** Ask ER or the MICU triage fellow for a **trialysis line**; admit to MICU. Ask the apheresis RN to **save the product in the FGR fridge** and coordinate with the research team. **Never in APL** |
| **Cytarabine** | **1 g ×1** may be considered for leukostasis when apheresis will be delayed |

## APL OVERNIGHT

- **Start ATRA on any concern for APL** — do not wait for PML::RARA
- **Smear:** bilobed / folded nuclei ± Auer rods. **Page pathology on call (12005)** if unsure
- **ATRA 45 mg/m²/day PO divided q12h** (= 22.5 mg/m² BID) — overnight you can give a one-time dose
- **DIC labs q6–8h.** Transfuse **platelets >30**, **fibrinogen >100**, **INR <2**

## REGIMEN DOSING

| Regimen | Dosing |
|---|---|
| **7+3** | Cytarabine **100–200 mg/m²/day** IV continuous infusion ×7 days from day 1 (**SHC uses 100**) + daunorubicin **60 mg/m²** IV daily **days 1–3** *schedule E1900; 60 vs 90 from AML17* |
| **7+3 + GO** | Add gemtuzumab ozogamicin **3 mg/m²** (max 4.5 mg) IV over 2 h on **days 1, 4, 7** *ALFA-0701 / Lambert 2019* |
| **7+3 + midostaurin** | Add midostaurin **50 mg PO q12h on days 8–21** *RATIFY / Stone 2017* |
| **CPX-351** | Liposomal daunorubicin **44 mg/m²** + cytarabine **100 mg/m²** IV on **days 1, 3, 5** *Lancet 2018* |
| **FLAG-IDA-VEN** | Fludarabine **30 mg/m²** + cytarabine **1.5–2 g/m²** IV days 2–6; idarubicin IV (**ND-AML 8 mg/m² days 4–6**; **R/R 6 mg/m² days 4–5**); filgrastim 5 mcg/kg days 1–7; venetoclax ramp-up (100/200/400 mg) days 1–14 *DiNardo 2021* |
| **CLAD** / **LDAC** / **VEN** | Cladribine **5 mg/m²** IV days 1–5 + cytarabine **20 mg/m²** (SQ BID days 1–10; **20 mg/m² IV daily preferred at SHC**) + venetoclax PO days 1–21, alternating with azacitidine *Kadia 2022* |
| **MEC** | Mitoxantrone **6–8 mg/m²** + etoposide **80–100 mg/m²** + cytarabine **1 g/m²**, all daily **days 1–5** (some series run 6 days) — dosing varies by protocol, check yours *relapsed / refractory* |
| **CLAG-M** | Cladribine **5 mg/m²** IV days 1–5 + cytarabine **2 g/m²** IV days 1–5 + mitoxantrone **10 mg/m²** IV days 1–3 + G-CSF **300 mcg** days 0–5 *relapsed / refractory* |
| **HMA** / **VEN** | Venetoclax (ramp-up) daily ×28 days + **decitabine 20 mg/m² days 1–5** or **azacitidine 75 mg/m² days 1–7** *VIALE-A / DiNardo 2020* |

## APL INDUCTION — DOSING

| Regimen | Dosing |
|---|---|
| **ATO** / **ATRA** | Arsenic trioxide **0.15 mg/kg** IV over 2 h daily from day 1 + ATRA **22.5 mg/m² PO BID** from day 1 — both until hematologic CR or **max 60 days** *APL0406 / Lo-Coco 2013* |
| **ATO** / **ATRA** / **IDA** | ATO **0.15 mg/kg** IV days 9–36 + ATRA **22.5 mg/m² BID** days 1–36 + idarubicin on days 2, 4, 6, 8 by age: **≤60 yr 12 mg/m²** · **61–70 yr 9 mg/m²** · **>70 yr 6 mg/m²** *APML4 / Iland 2012* |
| **ATO** / **ATRA** / **GO** | ATO **0.3 mg/kg** IV daily days 1–5, then **0.25 mg/kg twice weekly weeks 2–8** + ATRA **22.5 mg/m² BID** from day 1 until CR or max 60 days + gemtuzumab **6 mg/m²** IV once on day 1 **— high risk only (WBC >10k)** *AML17 APL / Burnett 2015* |